



Digital Medical Device Lifecycle & Risk Management Platform



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Project Overview

Problem Identification: Manual logs, non-transparent & delayed preventive maintenance of faults in medical devices

Objectives: Design a digital platform that enables hospitals to track, assess, and optimize the lifecycle of medical devices while aligning with ISO 14971 risk management standards.

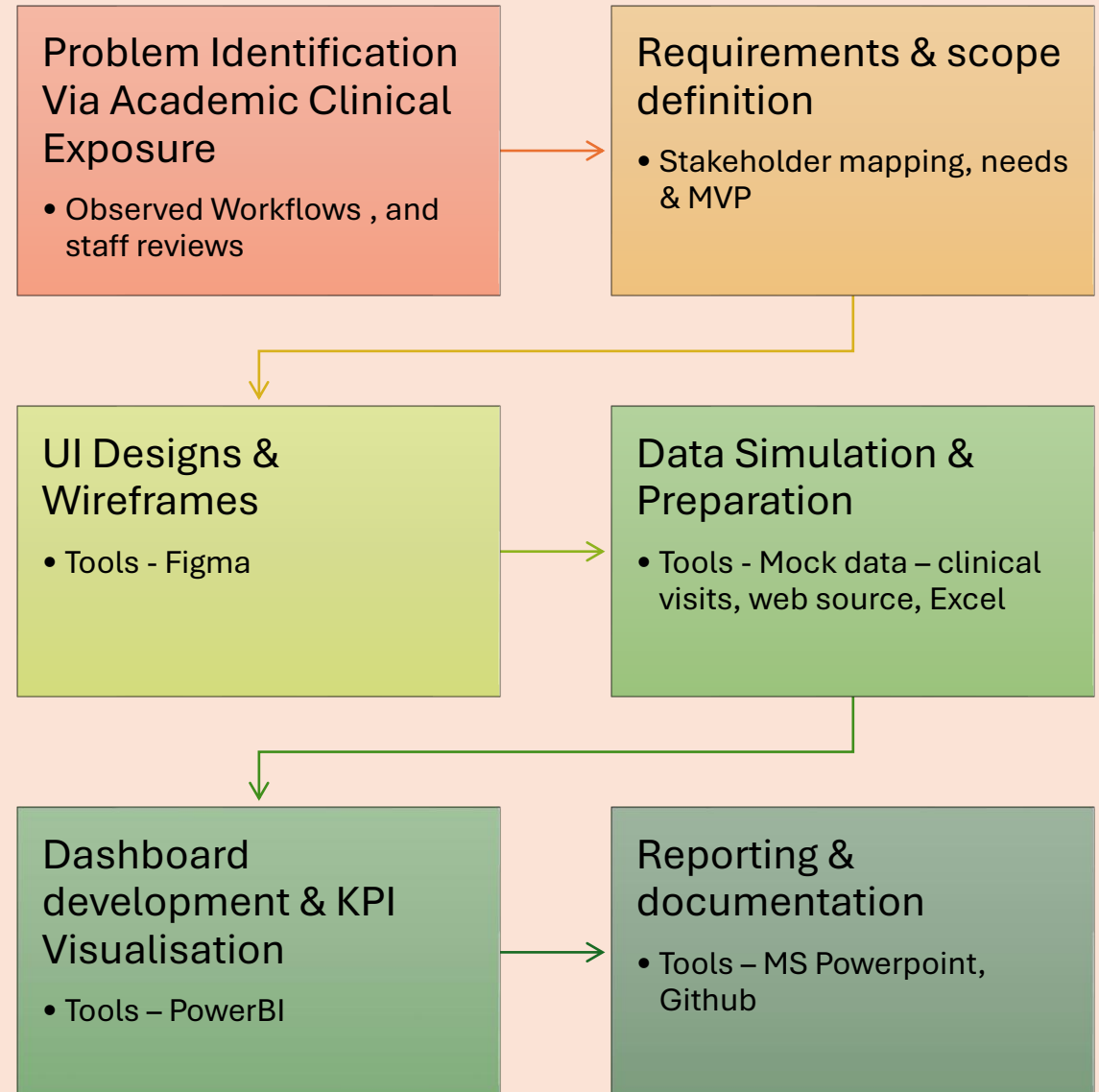
Goals

- Track device health and usage
- Visualize risks and maintenance load
- Create audit-ready reports

Location: Large Government Hospital

Key metrics tracked: Device Status, Equipment Type, Location, Technician, Trends

Project Plan & Methodology



Stakeholder Mapping



Stakeholder Role	Responsibility	Pain Points	Needs / Expectations
Biomedical Engineer	Maintenance, calibration, inventory tracking	Paper logs, lack of alerts, no failure analytics	Real-time dashboards, performance insights
Clinical Staff (Nurses)	Device operation, error reporting	Slow reporting, poor training on new devices	Easy reporting app, training materials, clear SOPs
Maintenance Technician	Field repairs, emergency response	Doesn't know what to fix until it breaks	Preventive maintenance forecast, auto-alerts
Hospital Admin	Budgeting, compliance, procurement	No lifecycle cost tracking, weak regulatory visibility	KPIs dashboard, risk flags, asset retirement roadmap
Regulatory Officer	Ensuring ISO 14971/NABH compliance	Manual logs, difficult audit prep	Pre-built risk reports, auto-documentation of failures
Pharmacist	Cold storage, drug-device interaction monitoring	Alerts for cold chain failure, lack of error incident visibility	Device health notifications, integration with medication logs

Competitor Analysis



Key Functionality/Need	GE CARESCAPE	AgileMD	Indian Govt Hospitals	Proposed Platform
Real-time device inventory	✓	✗	✗	✓
ISO 14971-compliant risk scoring	✗	✗	✗	✓
Incident reporting (mobile/web)	✗	✓	✗	✓
Preventive maintenance alerts	✗	✗	✗	✓
Dashboard with KPIs	✓	✗	✗	✓
Auto-generated compliance reports	✗	✗	✗	✓
Stakeholder-specific access levels	✓	✗	✗	✓
Lifecycle cost tracking	✓	✗	✗	✓



User Personas

Persona	Role in Hospital Workflow	Pain Points	Goals / What Success Looks Like
AJ Biomedical Engineer	Maintains and tracks all medical devices	Manual logs, no risk score visibility, audit overload	Real-time dashboard, automated risk scoring
Ni Head Nurse	Uses devices and reports issues	Slow incident reporting, poor feedback loop	One-click reporting, status tracking on incidents
Hospital Admin Executive (ops & Compliance)	Handles compliance, budget, hospital ops	Lacks risk dashboards, can't track utilization or costs	KPI dashboards, cost-risk analytics, audit-ready reports
Rx Chief Pharmacist	Manages cold-chain and drug-linked devices	No alerts for device failure, pharmacy losses	Cold chain alerts, audit trail for storage devices

Unique Value Proposition

A lightweight, audit-ready platform for large hospitals to track medical device health, automate risk scoring, and streamline compliance—without needing expensive hardware or infrastructure.

Minimum Viable Product

Feature Name	Description
Device Inventory Tracker	Register devices with department, model, status, and maintenance dates
Incident Reporting Module	Allow clinical staff to report issues via web or mobile app
ISO 14971 Risk Scoring Tool	Auto-calculate risk score based on severity, likelihood, and detectability
Maintenance Log & Alerts	Trigger maintenance tasks based on risk score or scheduled dates
Admin Dashboard	View KPIs: incidents per dept, risk trends, device utilization
Compliance Report Generator	Export PDF/Excel of incidents, risk actions for audit



MoSCoW Prioritization Framework

Feature Name	BM Engg.	Nurse	Rx Pharmacist	Admin	MoSCoW Category	Justification
Device Inventory Management	✓	✗	✓	✓	Must Have	Central to tracking and managing all medical devices.
Risk Scoring (ISO 14971)	✓	✗	✓	✓	Must Have	Essential for compliance and prioritizing maintenance.
Incident Reporting (Mobile/Web)	✓	✓	✓	✗	Must Have	Enables timely reporting and resolution of device issues.
Maintenance Scheduling	✓	✗	✗	✓	Should Have	Important for planning and reducing device downtime.
Dashboard – Real-Time KPIs	✓	✗	✓	✓	Must Have	Provides stakeholders with immediate insights into system status.
Compliance Report Generator	✓	✗	✓	✓	Must Have	Necessary for audits and regulatory compliance.
Cold Storage Alerts	✗	✗	✓	✗	Should Have	Critical for pharmacists to ensure medication safety.
Multi-Role Access Control	✓	✓	✓	✓	Should Have	Ensures appropriate access levels across different user roles.
Device Utilization Analytics	✓	✗	✗	✓	Could Have	Helps in optimizing device usage and budgeting.
Mobile App for Nurses	✗	✓	✗	✗	Could Have	Enhances mobility and ease of use for nursing staff.
AI-Powered Predictive Maintenance	✓	✗	✗	✓	Won't Have	Advanced feature for future consideration to predict device failures.

Wireframes

Head Nurse's dashboard Incident Initiator

Incident Report FormNI Nurse
Logout

[\[Report New Incident\]](#) [\[View My Reports\]](#) [\[Help/SOPs\]](#)

Select Device

Search and Select

Location

Select Ward/room

Issue

power failure, inaccuracy, noise, etc

Additional details

Add details about incident (photos, videos)

Submit

Cancel

Confirmation : Incident #ID logged BME dept. notified

[\[Track Status\]](#) [\[Report Another Issue\]](#)

Rx Pharmacist's dashboard Asset specific stakeholder

Pharmacy Cold Chain & inventoryMonitoring
Rx Cheif PharmacistLogout

Live Cold Chain Overview

Vaccine Fridge #1: 2.3°C ✓
Fridge #2 (Biologics): 7.8°C ▲ (High)
[| \[View Graph\]](#) [\[Acknowledge Alert\]](#)

Inventories at Risk

Insulin Lot # Z56 - 70 units - Cold chain breach reported
Toxoids & Vaccines Batch T56 - 16 VIALS - EXPIRING in 4 days
Inventory Supply Status : Purchase order passed for 256 vials
Vaccines, other medicines delivered on 7 Apr, 2025

Cold chain History

Line chart showing teperature readings over past 7 days with outlier readings highlighted RED
[| \[Download report\]](#)



Compliance Documentation

WHO PQS Guidelines: Linked
Daily logs – Auto-filled ✓

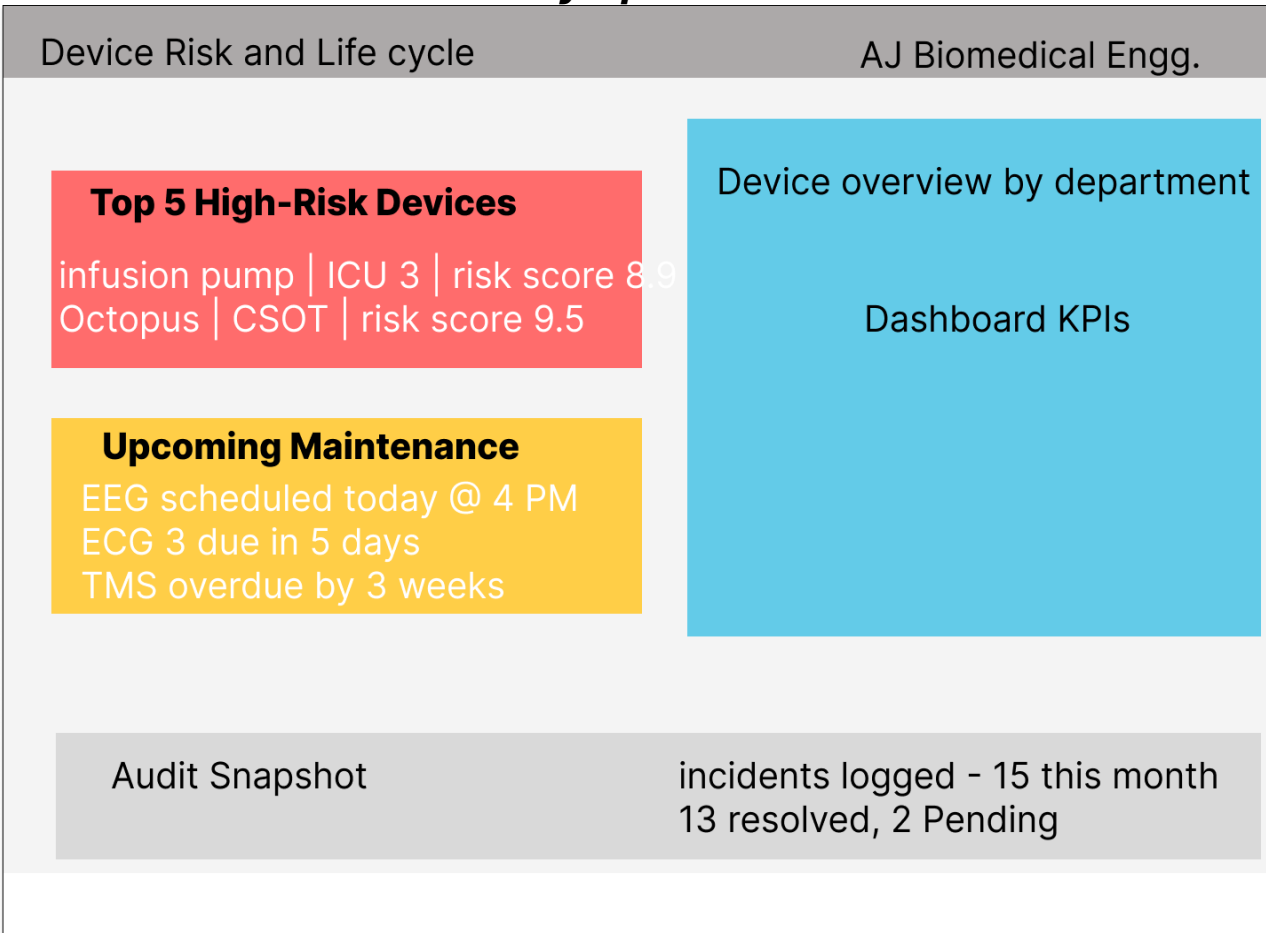
Complaints

Log Complaints

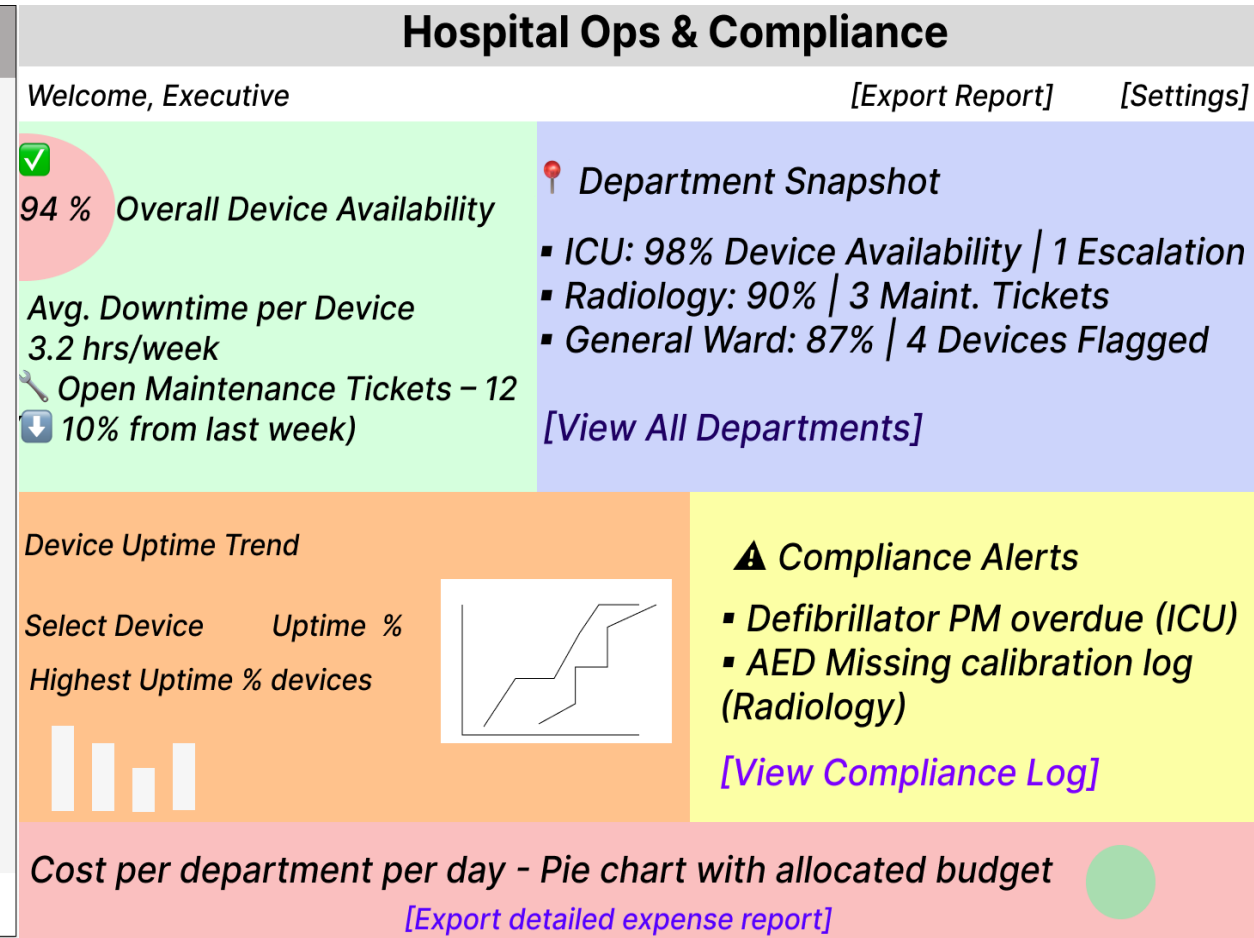
[\[Export report\]](#) [\[Safety Report\]](#)

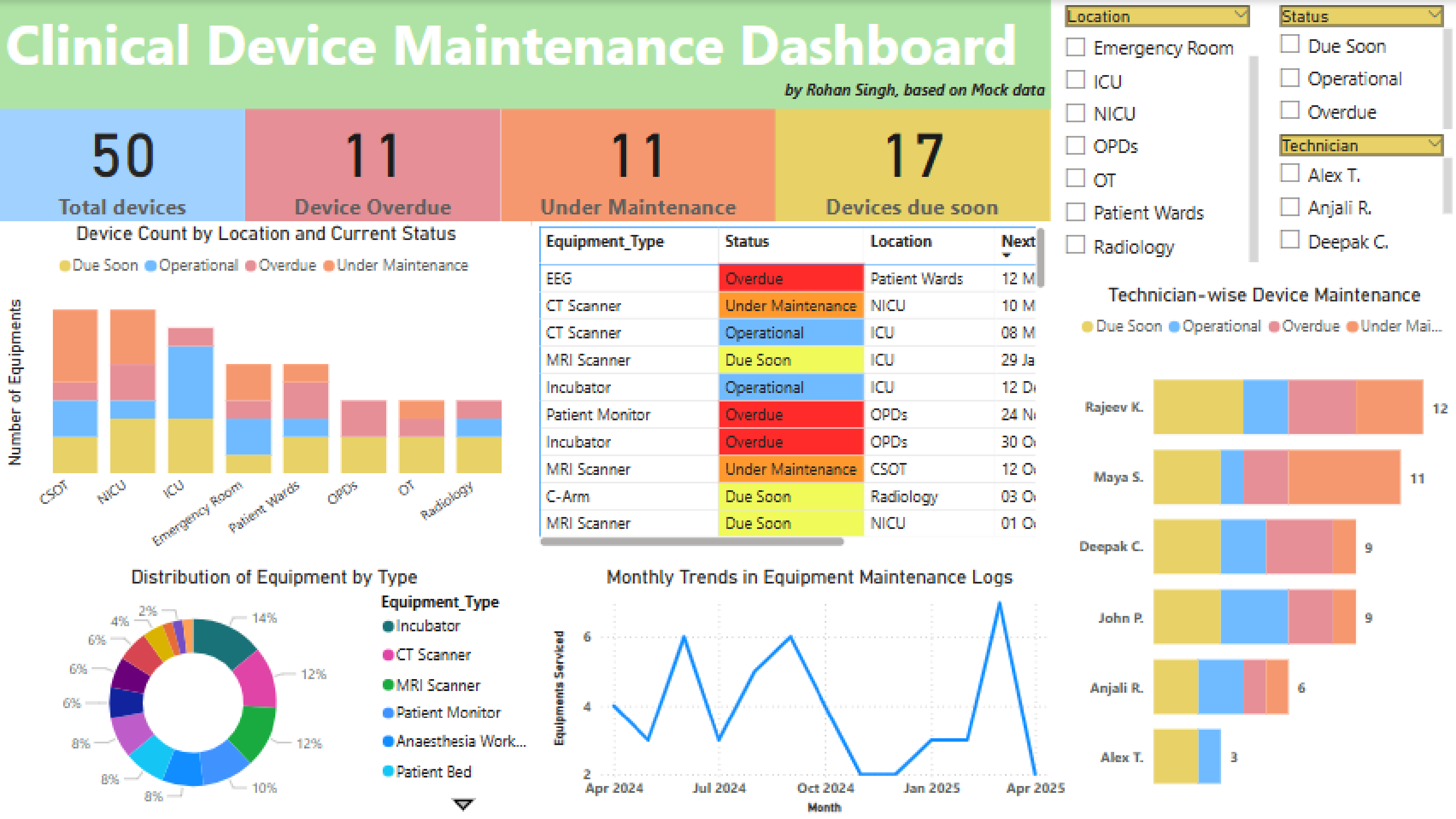
Wireframes

AJ Biomedical Engineer's dashboard Daily operator

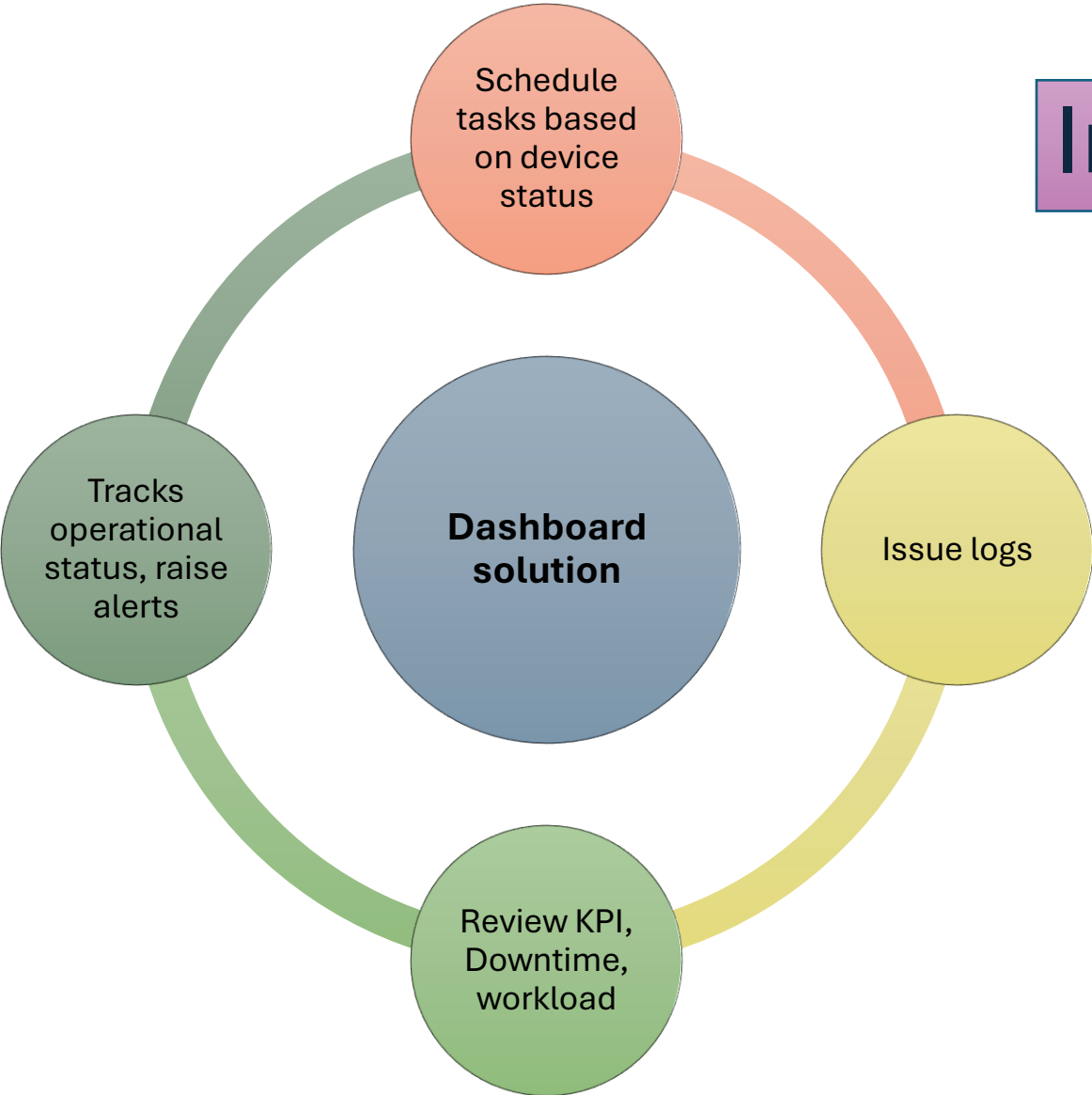


Hospital Admin executive dashboard Decision Maker





Insights



Device Risk & Maintenance

Location Based Status



Technician Overload

Equipment Distribution



Trend Analysis

Audit Readiness



Outcome & Benefits

- Visual system to alert risks before failure
- Better tech allocation
- Simplified audits for NABH

Future Scope:

- Integrate with IoT / real-time alerts
- Mobile technician interface
- Predictive maintenance with AI

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